

A Gentle Introduction to Inverse Reinforcement Learning

What Must Be Assumed, and What Must Be Approximated

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Abstract

Forward reinforcement learning is a well-posed problem: a reward determines an optimal value function uniquely, and a contraction mapping computes it. Inverse reinforcement learning is not well-posed, and it fails in two independent ways at once. It fails *statistically*, because infinitely many rewards explain the same behaviour, so no amount of data can single one out without extra assumptions. And it fails *computationally*, because every principled way of scoring a candidate reward requires an expectation under the policy that reward induces—an entire forward RL problem nested inside every gradient step.

These notes use that pair as their spine. Each method is presented as a specific answer to two separate questions: *what did you assume in order to make the answer unique?* and *how did you compute the inner expectation?* Read this way, the sequence Max-Margin $\rightarrow$ MaxEnt $\rightarrow$ GAIL $\rightarrow$ AIRL stops looking like a list of tricks and becomes a single argument.

We develop Maximum-Entropy IRL first in the finite, known-model setting, where the partition function is an honest finite sum and can be computed exactly by a soft Bellman backward pass. Only after the reader can point at the two places where $P(s' | s, a)$ is consumed do we ask what to do without it. Part [IV](#) then treats, in order, the three questions that trouble almost every beginner: what changes when the problem stops being finite, whether the reward features must be hand-crafted, and what remains possible when the transition kernel is unknown. We are deliberately careful about four points that are frequently stated incorrectly in secondary sources: the maximum-entropy objective requires a dynamics-induced reference measure from its very first line; naive maximum entropy is risk-seeking under stochastic dynamics and must be replaced by maximum *causal* entropy; GAIL’s discriminator carries *no* reward information at convergence; and AIRL’s transfer guarantee requires a *state-only* reward term.

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How to Read These Notes

A first course in IRL usually presents four algorithms in historical order and leaves the reader to infer why each one replaced the last. These notes instead fix two coordinates and locate every method on them.

The Two Coordinates

Coordinate 1 — The identification problem: *what must be assumed?* Behaviour does not determine reward. The forward map $R \mapsto \pi^*$ destroys information, and no estimator can recreate it. Every IRL method therefore smuggles in (i) a *behaviour model* saying how reward produces action, and (ii) a *selection principle* choosing one representative from the surviving equivalence class, and (iii) a *reward class* restricting what R may look like. These are modelling commitments, not discoveries. A method is only as trustworthy as they are.

Coordinate 2 — The computation problem: *what must be approximated?* Every selection principle scores a candidate reward R by comparing the expert’s behaviour statistics against those of the policy that R induces. Producing that second quantity means solving a forward RL problem *inside* the loop, and then taking an expectation under its solution. In a finite known MDP both steps are exact. Outside that setting they are not, and the history of IRL is largely the history of progressively cheaper surrogates for that one expectation.

The two coordinates are genuinely independent, and keeping them apart resolves most early confusion. A method can be statistically honest and computationally hopeless (exact tabular MaxEnt in a large MDP), or computationally excellent and statistically silent about reward (GAIL). When someone says “IRL is ill-posed,” ask which coordinate they mean.

The route

- Part [I](#) establishes the forward direction and, more importantly, isolates the four properties that make it well-posed, so that we can watch each one fail in turn.
- Part [II](#) attacks Coordinate 1 head-on: what precisely can and cannot be inverted, and what the honest target of IRL therefore is.
- Part [III](#) develops Max-Margin and Maximum-Entropy IRL in the one regime where Coordinate 2 costs nothing—finite spaces, known P . This is the reference point for everything after it. The partition function is a finite sum; we compute it two ways and check the numbers agree.
- Part [IV](#) leaves that regime along its three natural exits, which are exactly the three questions beginners ask: finite $\rightarrow$ infinite, given $\rightarrow$ learned features, model-based $\rightarrow$ model-free.
- Part [V](#) shows GAIL and AIRL are the answers those three exits force on us, not an unrelated import from generative modelling.
- Part [VI](#) puts every method back on the two coordinates in a single table.

Each method section closes with a **Ledger** box recording, in the method’s own terms, what it assumed and what it approximated. If you remember nothing else from these notes, remember to fill in that ledger for any IRL paper you read.

Notation

Symbols are reused across the IRL literature with maddening inconsistency. We fix a single convention here and keep it for the whole document. In particular: P always denotes the transition kernel and never a distribution over trajectories (those are lower-case p); the discount is always γ and never λ ; ϕ is always a per-step feature and Φ always its discounted trajectory sum; the entropy temperature is β ; and the margin variable is ξ , so that t is free to mean time and k to mean an iteration index.

Table 1: Notation used throughout.

Symbol	Meaning	Type / definition
<i>The environment</i>		
$\mathcal{S}, \mathcal{A}$	State and action spaces	finite unless stated otherwise
$P(s' s, a)$	Transition kernel	$\mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$
$\mathbf{P}^\pi$	State-to-state matrix induced by π	$\in \mathbb{R}^{ \mathcal{S} \times \mathcal{S} }$, row-stochastic
$d_0(s)$	Initial state distribution	$\in \Delta(\mathcal{S})$
γ	Discount factor	$\gamma \in (0, 1)$
T	Horizon (finite-horizon sections only)	$T \in \mathbb{N}$
$\mathcal{M}$	MDP without reward, $(\mathcal{S}, \mathcal{A}, P, \gamma, d_0)$	the “dynamics”
<i>Reward, policy, value</i>		
$R(s, a)$	Reward function	$\mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, $ R \leq R_{\max}$
$R_w(s, a)$	Reward in a parametric class	e.g. $R_w(s, a) = w^\top \phi(s, a)$
$\pi(a s)$	Stationary policy	$\mathcal{S} \rightarrow \Delta(\mathcal{A})$
π_E	Expert policy	assumed (near-)optimal for R^*
V^π, Q^π	Value functions under π	$\mathcal{S} \rightarrow \mathbb{R}$, $\mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$
V^*, Q^*	Optimal value functions	fixed points of $\mathcal{B}^*$
$V_{\text{soft}}, Q_{\text{soft}}$	Soft (entropy-regularised) value functions	fixed points of $\mathcal{B}_{\text{soft}}$
$\mathcal{B}^*, \mathcal{B}_{\text{soft}}$	Bellman optimality / soft operators	$\mathbb{R}^{ \mathcal{S} } \rightarrow \mathbb{R}^{ \mathcal{S} }$
$J_R(\pi)$	Expected discounted return of π under R	$\mathbb{E}_\pi[\sum_t \gamma^t R(s_t, a_t)]$
β	Entropy temperature / regularisation weight	$\beta > 0$
$\Psi(s)$	Potential function for reward shaping	$\mathcal{S} \rightarrow \mathbb{R}$
<i>Trajectories and data</i>		
τ	Trajectory	element of $\mathcal{T}$
$\mathcal{T}$	Set of feasible trajectories	finite when $\mathcal{S}, \mathcal{A}, T$ finite
$\mathcal{D}_E$	Expert dataset $\{\tau^{(1)}, \dots, \tau^{(m)}\}$	finite demonstrations
$p_\pi(\tau)$	Trajectory distribution induced by π	$d_0 \prod_t \pi P$
$q(\tau)$	Dynamics-induced reference measure	$d_0(s_0) \prod_t P(s_{t+1} s_t, a_t)$
<i>Features and occupancy</i>		
$\phi(s, a)$	Per-step feature map	$\mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$
$\Phi(\tau)$	Discounted trajectory feature	$\sum_{t \geq 0} \gamma^t \phi(s_t, a_t) \in \mathbb{R}^d$
$\mu(\pi)$	Feature expectation of π	$\mathbb{E}_{\tau \sim p_\pi}[\Phi(\tau)]$
μ_E	Empirical expert feature expectation	$\frac{1}{m} \sum_i \Phi(\tau^{(i)})$
$\rho_\pi(s, a)$	Discounted occupancy measure	$(1 - \gamma) \sum_t \gamma^t \Pr_\pi(s_t = s, a_t = a)$
ρ_E	Expert occupancy measure	estimated from $\mathcal{D}_E$
w	Reward weight vector	$w \in \mathbb{R}^d$
<i>Maximum entropy</i>		
$\mathcal{H}_q(p)$	Entropy of p relative to q	$-\sum_\tau p(\tau) \log \frac{p(\tau)}{q(\tau)}$
$Z(w)$	Partition function	$\sum_{\tau \in \mathcal{T}} q(\tau) e^{w^\top \Phi(\tau)}$
c	Lagrange multiplier for normalisation	$c \in \mathbb{R}$
ξ	Max-margin slack/margin variable	$\xi \in \mathbb{R}$
<i>Adversarial methods</i>		

Table 1: Notation used throughout.

Symbol	Meaning	Type / definition
$D(s, a)$	Discriminator; $\Pr[\text{pair came from expert}]$	$\mathcal{S} \times \mathcal{A} \rightarrow (0, 1)$
$g_\theta(s)$	AIRL task-reward network (state-only)	$\mathcal{S} \rightarrow \mathbb{R}$
$h_\omega(s)$	AIRL shaping network	$\mathcal{S} \rightarrow \mathbb{R}$
$f_{\theta, \omega}$	AIRL discriminator logit	$g_\theta(s) + \gamma h_\omega(s') - h_\omega(s)$
ψ, ψ^*	Reward regulariser and its convex conjugate	$\psi : \mathbb{R}^{\mathcal{S} \times \mathcal{A}} \rightarrow \overline{\mathbb{R}}$

Part I

The Well-Posed Direction

1 Forward Reinforcement Learning

Our purpose in this part is not to teach RL from scratch but to isolate *why* the forward problem is easy, in a form precise enough that we can later watch each reason fail.

1.1 Markov decision processes

Definition 1.1 (Markov Decision Process). A finite MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma, d_0)$ where $\mathcal{S}$ and $\mathcal{A}$ are finite sets, $P(s' | s, a)$ is a transition kernel, $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is bounded by $R_{\max}$, $\gamma \in (0, 1)$, and $d_0 \in \Delta(\mathcal{S})$. We write $\mathcal{M}$ for the reward-free part $(\mathcal{S}, \mathcal{A}, P, \gamma, d_0)$ and call it the *dynamics*.

The separation of $\mathcal{M}$ from R is not cosmetic. IRL is the problem of recovering R *given* $\mathcal{M}$ (or given the ability to interact with it) and given behaviour. Everything hinges on how much of $\mathcal{M}$ we are allowed to know.

For a stationary policy π ,

$$V^\pi(s) \triangleq \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid s_0 = s \right], \quad Q^\pi(s, a) \triangleq R(s, a) + \gamma \sum_{s' \in \mathcal{S}} P(s' | s, a) V^\pi(s'), \quad (1)$$

and the scalar objective is $J_R(\pi) \triangleq \mathbb{E}_{s_0 \sim d_0} [V^\pi(s_0)]$.

1.2 Policy evaluation is a linear solve

Fix π . Define $\mathbf{V}^\pi \in \mathbb{R}^n$ with $n = |\mathcal{S}|$, the induced reward vector $[\mathbf{R}^\pi]_i = \sum_a \pi(a | s_i) R(s_i, a)$, and the induced transition matrix $[\mathbf{P}^\pi]_{ij} = \sum_a \pi(a | s_i) P(s_j | s_i, a)$. The Bellman expectation equation is then linear:

$$\mathbf{V}^\pi = \mathbf{R}^\pi + \gamma \mathbf{P}^\pi \mathbf{V}^\pi. \quad (2)$$

Theorem 1.2 (Exact policy evaluation). For $\gamma \in (0, 1)$ and row-stochastic $\mathbf{P}^\pi$, the matrix $\mathbf{I} - \gamma \mathbf{P}^\pi$ is invertible and

$$\mathbf{V}^\pi = (\mathbf{I} - \gamma \mathbf{P}^\pi)^{-1} \mathbf{R}^\pi = \sum_{k=0}^{\infty} \gamma^k (\mathbf{P}^\pi)^k \mathbf{R}^\pi. \quad (3)$$

Proof. Since $\mathbf{P}^\pi$ is row-stochastic, $\|\mathbf{P}^\pi\|_\infty = 1$, hence $\|\gamma \mathbf{P}^\pi\|_\infty = \gamma < 1$. The Neumann series $\sum_{k=0}^{\infty} \gamma^k (\mathbf{P}^\pi)^k$ therefore converges absolutely in the induced ∞ -norm and equals $(\mathbf{I} - \gamma \mathbf{P}^\pi)^{-1}$. Applying it to (2) gives the claim. $\square$

Remark 1.3. The Neumann series has a reading we will need repeatedly: $(\mathbf{I} - \gamma \mathbf{P}^\pi)^{-1}$ is, up to normalisation by $(1 - \gamma)$, the matrix of *discounted state occupancies*. Value is an occupancy-weighted sum of rewards. That single observation is the bridge from classical IRL to GAIL, and we develop it properly in Section 5.9.

1.3 Optimality is a fixed point

Definition 1.4 (Bellman optimality operator). $(\mathcal{B}^* V)(s) \triangleq \max_{a \in \mathcal{A}} [R(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s')]$.

Theorem 1.5 (Contraction). $\|\mathcal{B}^*V_1 - \mathcal{B}^*V_2\|_\infty \leq \gamma\|V_1 - V_2\|_\infty$. Consequently $\mathcal{B}^*$ has a unique fixed point V^* , value iteration converges geometrically, $\|V^{(k)} - V^*\|_\infty \leq \gamma^k\|V^{(0)} - V^*\|_\infty$, and any greedy policy with respect to V^* is optimal.

Proof. For each s , $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$; apply this with $f(a) - g(a) = \gamma \sum_{s'} P(s' | s, a)(V_1 - V_2)(s')$ and bound the sum by $\gamma\|V_1 - V_2\|_\infty$ using $\sum_{s'} P = 1$. Banach’s fixed-point theorem gives existence, uniqueness and the rate. $\square$

1.4 Why the forward direction is easy, in four bullets

The Four Pillars of Forward RL

1. **The unknown is finite-dimensional and the equations are exact.** $V^* \in \mathbb{R}^{|S|}$ solves a finite system.
2. **The solution operator is a contraction.** Uniqueness and a geometric convergence rate come for free.
3. **Expectations are finite sums over a known kernel.** Every $\sum_{s'} P(s' | s, a)(\cdot)$ can be evaluated exactly.
4. **The objective is given.** There is a ground truth against which any answer can be scored.

Pillar 4 is what IRL removes, and that is Coordinate 1. Pillars 1–3 are what large-scale or unknown environments remove, and that is Coordinate 2. It is worth noticing that forward RL already has a well-developed answer to losing pillars 1–3, which we now recall in a single page, because IRL inherits it wholesale.

1.5 Model-based and model-free forward RL

1. **Model-based.** P and R are known. Solve by (3), or by value/policy iteration. Value iteration converges linearly with factor γ ; policy iteration terminates finitely (at most $|\mathcal{A}|^{|S|}$ policies) and is typically far faster in practice.
2. **Model-free.** P is unknown; the agent may only sample transitions. Temporal-difference methods (Q -learning, SARSA) replace the exact expectation $\sum_{s'} P(\cdot)$ by sampled successors. Tabular Q -learning converges almost surely to Q^* under the Robbins–Monro conditions $\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$ with every (s, a) visited infinitely often [18, 19]. With function approximation—policy-gradient methods included [22]—the combination of bootstrapping, off-policy sampling and approximation—the *deadly triad*—can destroy the contraction, and neither uniqueness nor convergence is guaranteed.

Remark 1.6 (Ill-conditioning as $\gamma \rightarrow 1$). In the ∞ -norm, $\kappa_\infty(\mathbf{I} - \gamma\mathbf{P}^\pi) \leq \frac{1+\gamma}{1-\gamma}$, so the linear solve degrades as $\gamma \rightarrow 1^-$. The same $(1 - \gamma)^{-1}$ factors reappear in sample-complexity bounds: obtaining an ε -optimal policy with a generative model costs $\tilde{O}(\frac{|S||\mathcal{A}|}{\varepsilon^2(1-\gamma)^3})$ samples [21]. The effective horizon $(1 - \gamma)^{-1}$ is the quantity that hurts, in every regime, forward and inverse alike.

Part II

The Ill-Posed Direction

2 Setting Up the Inverse Problem

In many domains—driving, surgery, aerobatic flight, dialogue—writing down a reward is harder than demonstrating the behaviour. Hand-specified rewards are routinely mis-specified, and optimising a mis-specified reward produces the failure mode known as reward hacking. IRL proposes to infer the objective from behaviour instead.

The Two Directions			
Forward RL:	dynamics $\mathcal{M}$ + reward R	$\xrightarrow{\text{optimise}}$	optimal policy π^*
Inverse RL:	dynamics $\mathcal{M}$ + demonstrations $\mathcal{D}_E$	$\xrightarrow{\text{infer}}$	reward R (then, usually, π)

We observe m trajectories $\mathcal{D}_E = \{\tau^{(1)}, \dots, \tau^{(m)}\}$, with $\tau = (s_0, a_0, s_1, a_1, \dots, s_T)$, and we define the discounted trajectory return $R(\tau) \triangleq \sum_{t \geq 0} \gamma^t R(s_t, a_t)$.

Assumption 2.1 (Expert (near-)optimality). $\mathcal{D}_E$ was generated by a policy π_E that is optimal, or near-optimal, for some unknown reward R^* in the given dynamics $\mathcal{M}$.

Assumption 2.1 is the *behaviour model* in its crudest form. Taken literally it yields only inequalities:

$$J_R(\pi_E) \geq J_R(\pi) \quad \text{for all } \pi \in \Pi. \quad (4)$$

A system of inequalities is a weak thing to estimate from. Much of the technical content of modern IRL consists of replacing (4) by a *probabilistic* behaviour model, which turns inequalities into a likelihood and therefore into something a gradient can act on. We will see exactly that happen in Section 5.

3 What Exactly Can Be Inverted?

Calling IRL “inverse optimisation” invites the wrong intuition, because ordinary inversion presumes injectivity. The forward map

$$R \mapsto \Pi^*(R) = \{\text{policies optimal for } R\} \quad (5)$$

is drastically many-to-one. Before choosing an algorithm we should know precisely what information the map destroys.

Three Objects That Must Not Be Confused	
1.	V^* is unique, given $\mathcal{M}$ and R : it is the fixed point of a contraction.
2.	π^* need <i>not</i> be unique: ties in the greedy step give a set $\Pi^*(R)$.
3.	R is <i>far</i> from unique, given behaviour: Π^* has enormous fibres.
Uniqueness of the forward solution says nothing whatever about uniqueness of the inverse. These three statements are about three different maps.	

3.1 The solution set is a polyhedral cone

Ng and Russell gave the exact characterisation for a finite MDP and a fixed deterministic policy [1].

Theorem 3.1 (Characterisation of consistent rewards, Ng & Russell 2000). *Let $\mathcal{S}$ be finite, let $R \in \mathbb{R}^{|\mathcal{S}|}$ be a state-only reward, and let $\mathbf{P}_a$ denote the matrix with entries $P(s' | s, a)$. A deterministic policy π , which we may relabel so that $\pi(s) \equiv a_1$, is optimal for R if and only if*

$$(\mathbf{P}_{a_1} - \mathbf{P}_a)(\mathbf{I} - \gamma \mathbf{P}_{a_1})^{-1} R \succeq 0 \quad \text{for all } a \in \mathcal{A} \setminus \{a_1\}. \quad (6)$$

Two consequences are immediate and worth dwelling on.

Corollary 3.2. *The set of rewards consistent with an observed policy is a closed convex cone: it is closed under non-negative scaling and under addition. In particular it always contains $R \equiv 0$, for which every policy is optimal, so observing behaviour alone can never exclude the degenerate explanation “nothing matters”.*

Sketch of the degeneracies. Beyond $R \equiv 0$: any constant $R \equiv c$ makes all policies optimal; and for any potential $\Psi : \mathcal{S} \rightarrow \mathbb{R}$, the shaped reward

$$R'(s, a, s') = R(s, a) + \gamma \Psi(s') - \Psi(s) \quad (7)$$

satisfies $J_{R'}(\pi) = J_R(\pi) - \mathbb{E}_{s_0 \sim d_0}[\Psi(s_0)]$ for every π , so it shifts all policy values by the same constant and preserves the entire preference ordering over policies—hence $\Pi^*(R') = \Pi^*(R)$. $\square$

Remark 3.3 (Correct attribution). The optimality-preserving property of potential-based shaping (7) is due to Ng, Harada and Russell [2], not the 2000 IRL paper; that work also proves the converse, that potential-based shaping is the *only* form of additive reward modification that preserves optimality for *all* transition kernels. This converse is exactly why potential shaping is the right notion of “nuisance parameter” in IRL, and it is the ambiguity that AIRL sets out to quotient away in Section 11.

3.2 A minimal example

Consider a single state and a single decision between **left** and **right**; the expert chooses **left**. All we learn is

$$R(s, \text{left}) \geq R(s, \text{right}). \quad (8)$$

The pairs $(1, 0)$, $(10, -3)$ and $(0, -100)$ are equally consistent. One observation cannot pin down two numbers, and no amount of repetition of the same observation will help: additional demonstrations add inequalities, which sharpen the *preference* structure but never fix an absolute scale. Notice also that this example is already statistically informative in one respect—it does rule out $R(s, \text{left}) < R(s, \text{right})$. IRL is not hopeless; it is *partially identifying*.

3.3 The honest target

Given all this, the scientifically defensible outputs of IRL are:

1. an **equivalence class** of rewards inducing the observed behaviour;
2. a **single representative** chosen by an explicit selection principle (maximum margin, maximum entropy, a Bayesian prior, sparsity);
3. or simply a **policy** whose occupancy measure matches the expert’s, with no claim about reward at all.

Item 3 is imitation learning, and it is often what one actually wants. Item 2 is what most IRL papers deliver. Item 1 is what the data actually supports. Confusing 2 with 1 is the single most common overclaim in the literature; recent work makes the partial-identifiability picture precise and shows that even with infinite data and a known model, the reward is only determined up to explicitly characterisable transformations [16, 17].

A claim to be suspicious of

“Our method recovers the true reward function.” No method recovers a reward function from behaviour alone. Methods recover a representative of an equivalence class, selected by an assumption the paper made. The useful question is never “is the reward right?” but “is the reward right *for the use I have in mind?*” — and the two most common uses, reproducing behaviour in the same MDP and transferring behaviour to a different MDP, need very different amounts of identification. Section 11 is entirely about the gap between them.

3.4 Choosing a reward class

The unknown $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is a function; if $\mathcal{S} \times \mathcal{A}$ is continuous it is an infinite-dimensional object. A finite vector $w \in \mathbb{R}^d$ appears only after we *choose* a model class, most commonly the linear one

$$R_w(s, a) = w^\top \phi(s, a). \quad (9)$$

This is a modelling commitment on Coordinate 1, not a law of IRL. Alternatives include kernel and Gaussian-process rewards, neural rewards R_θ , and reward classes defined implicitly by a discriminator architecture. We take up the question of *whether ϕ must be hand-designed* in full in Section 7; for now we simply record what (9) buys, because the next two sections lean on it entirely.

Proposition 3.4 (Linearity collapses return into a finite statistic). *Under (9),*

$$J_{R_w}(\pi) = \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t w^\top \phi(s_t, a_t) \right] = w^\top \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t \phi(s_t, a_t) \right] = w^\top \mu(\pi). \quad (10)$$

The vector $\mu(\pi) \in \mathbb{R}^d$ is a *sufficient statistic* of the policy for the purpose of evaluating any reward in the class. Everything else about π is irrelevant to w . This is the algebraic fact that makes classical IRL tractable, and losing it is the price of leaving the linear class.

Part III

Two Selection Principles, Computed Exactly

Throughout this part we stay in the *safe corner*: $\mathcal{S}$ and $\mathcal{A}$ finite, the kernel P fully known, and ϕ fixed in advance. Coordinate 2 therefore costs us nothing—every expectation below is a finite sum we can actually evaluate—and we can concentrate wholly on Coordinate 1. This is the only part of the notes in which every number can be checked by hand, and it is worth the patience: the objects defined here reappear, in approximated form, in every method that follows.

4 Selection Principle I: Maximum Margin

Abbeel and Ng [3] resolve the ambiguity of Section 3 with a geometric rule: among all rewards consistent with the data, prefer the one under which the expert beats the alternatives by the largest margin.

4.1 Feature expectations

Definition 4.1 (Feature expectation). $\mu(\pi) \triangleq \mathbb{E}_{\tau \sim p_\pi}[\Phi(\tau)] = \mathbb{E}_\pi[\sum_{t \geq 0} \gamma^t \phi(s_t, a_t)] \in \mathbb{R}^d$, and the empirical expert version is $\mu_E \triangleq \frac{1}{m} \sum_{i=1}^m \Phi(\tau^{(i)})$.

Lemma 4.2 (Feature matching suffices). *If $\|\mu(\pi) - \mu_E\|_2 \leq \varepsilon$ then for every w with $\|w\|_2 \leq 1$,*

$$|J_{R_w}(\pi) - w^\top \mu_E| = |w^\top (\mu(\pi) - \mu_E)| \leq \varepsilon \quad (11)$$

by Cauchy-Schwarz. Matching feature expectations therefore matches expected return simultaneously for every reward in the linear class, without knowing which one is the truth.

Lemma 4.2 is the reason apprenticeship learning can sidestep identification. It also states its own limitation with unusual clarity: the guarantee is quantified over $w \in \mathbb{R}^d$, and says nothing about rewards outside $\text{span}\{\phi_1, \dots, \phi_d\}$.

4.2 The optimisation

Because scaling $w \mapsto \alpha w$ scales any margin, a norm constraint is needed to make the problem bounded.

Max-Margin IRL (Abbeel & Ng, 2004)

$$\begin{aligned} & \max_{\xi \in \mathbb{R}, w \in \mathbb{R}^d} \quad \xi \\ & \text{subject to} \quad w^\top \mu_E \geq w^\top \mu(\pi_i) + \xi, \quad i = 0, \dots, k-1, \\ & \quad \quad \quad \|w\|_2 \leq 1. \end{aligned} \quad (12)$$

This is the hard-margin SVM in disguise, with μ_E the single positive example and $\{\mu(\pi_i)\}$ the negatives. Its dual is the minimum-distance problem

$$\min_{\bar{\mu} \in \text{conv}\{\mu(\pi_0), \dots, \mu(\pi_{k-1})\}} \|\mu_E - \bar{\mu}\|_2, \quad \text{with optimal } w \propto \mu_E - \bar{\mu}, \quad (13)$$

which is the geometric content of the algorithm: the reward direction always points from the best current imitation towards the expert. Solving (13) directly is Abbeel and Ng’s “projection” variant and avoids a QP solver.

4.3 The algorithm, and what it actually returns

Algorithm 1 Max-Margin / Apprenticeship Learning (Abbeel & Ng, 2004)

Require: dynamics $\mathcal{M}$, features ϕ , expert data $\mathcal{D}_E$, tolerance $\varepsilon > 0$

- 1: $\mu_E \leftarrow \frac{1}{m} \sum_{i=1}^m \sum_t \gamma^t \phi(s_t^{(i)}, a_t^{(i)})$
 - 2: pick any π_0 ; compute $\mu(\pi_0)$ exactly by solving $(\mathbf{I} - \gamma \mathbf{P}^{\pi_0})^{-1}$ once per feature coordinate
 - 3: **for** $k = 1, 2, \dots$ **do**
 - 4: (reward player) solve (12) over $\{\mu(\pi_0), \dots, \mu(\pi_{k-1})\}$ to get $(w^{(k)}, \xi^{(k)})$
 - 5: **if** $\xi^{(k)} \leq \varepsilon$ **then break**
 - 6: **end if**
 - 7: (policy player) $\pi_k \leftarrow$ exact forward RL solution for $R^{(k)} = (w^{(k)})^\top \phi$
 - 8: compute $\mu(\pi_k)$ exactly
 - 9: **end for**
 - 10: solve $\lambda^* = \arg \min_{\lambda \in \Delta_k} \|\mu_E - \sum_i \lambda_i \mu(\pi_i)\|_2$
 - 11: **return** the *mixed* policy $\tilde{\pi}$: sample $i \sim \lambda^*$ once at $t = 0$, then follow π_i forever
-

The output is a mixture, not a single policy

Algorithm 1 does *not* return the last π_k . Its guarantee is for the mixed policy $\tilde{\pi}$, and the reason is exactly (13): what is provably close to μ_E is a point in the *convex hull* of the $\mu(\pi_i)$, and convex combinations of feature expectations are realised by randomising over policies, not by any single one of them. This works because occupancy measures mix linearly— $\rho_{\tilde{\pi}} = \sum_i \lambda_i \rho_{\pi_i}$ —a fact we exploit again in Section 5.9. Presentations that return “the final policy” have quietly dropped the theorem.

Theorem 4.3 (Convergence, informal). *If $\phi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]^d$ then $\|\mu(\pi)\|_2 \leq \sqrt{d}/(1 - \gamma)$ for all π , and Algorithm 1 produces $\|\mu(\tilde{\pi}) - \mu_E\|_2 \leq \varepsilon$ after $k = O(\frac{d}{(1-\gamma)^2 \varepsilon^2} \log \frac{d}{(1-\gamma)\varepsilon})$ iterations—notably, a bound independent of $|\mathcal{S}|$. Combined with Lemma 4.2, $\tilde{\pi}$ then performs within ε of the expert under every reward in the class.*

4.4 Limitations, stated precisely

1. **Ties are broken arbitrarily.** Many (w, ξ) may achieve the same margin; the QP picks one. The selection principle is a convenience, not a discovery.
2. **Sub-optimal experts break it.** The constraints assert the expert is best. A noisy expert makes the feasible set shrink, or empty, and there is no likelihood to absorb the noise. This is precisely the gap maximum entropy fills.
3. **The inner solve is a full forward RL problem** per iteration. In the safe corner it is exact; outside it, it is the bottleneck.
4. **It is silent about which reward is right.** Its guarantee is about *performance*, quantified over the whole class.

Ledger: Max-Margin IRL

Assumed (Coordinate 1). Expert is exactly optimal; reward is linear in a fixed ϕ ; ties broken by maximal margin.

Approximated (Coordinate 2). Nothing, in the safe corner: $\mu(\pi)$ comes from an exact linear solve and the inner RL is exact. Cost is one full forward RL solve and one QP per iteration.

Delivers. A mixed policy matching μ_E , plus a reward direction w whose interpretation is weak.

5 Selection Principle II: Maximum Entropy

Max-margin IRL treats the expert as infallible. Real demonstrations are noisy, and worse, *systematically* noisy: an expert who is indifferent between two routes will take each about half the time, and a method built on strict inequalities has nothing to say about that. Ziebart et al. [4] replace the inequality model by a probabilistic one, and then select among the surviving models by maximum entropy, in the spirit of Jaynes’s original principle [6].

The setting for this section

Throughout Section 5 we take a **finite** $\mathcal{S}$ and $\mathcal{A}$, a **finite horizon** T , a **known** kernel P , and—following the original paper—we set $\gamma = 1$, writing $\Phi(\tau) = \sum_{t=0}^{T-1} \phi(s_t, a_t)$. Discounting and infinite horizons are restored in Section 6, where we explain why they are genuinely more delicate than simply inserting γ^t .

5.1 Why a distribution over trajectories, and over what?

We want a model $p(\tau)$ that (i) reproduces the expert’s feature statistics and (ii) commits to nothing else. The second requirement is what “maximum entropy” formalises. But entropy of *what*, exactly? This is where most treatments go wrong, so we go slowly.

A trajectory is produced by two agencies: the learner, who chooses actions, and the environment, who chooses successor states. If we maximise plain Shannon entropy, $-\sum_{\tau} p(\tau) \log p(\tau)$, we implicitly declare every trajectory equally plausible *a priori*, including trajectories the environment would essentially never produce. That is obviously wrong. The fix is to measure entropy relative to a reference measure encoding what the environment does.

Definition 5.1 (Dynamics-induced reference measure). For $\tau = (s_0, a_0, \dots, s_T)$ define

$$q(\tau) \triangleq d_0(s_0) \prod_{t=0}^{T-1} P(s_{t+1} \mid s_t, a_t). \quad (14)$$

Remark 5.2. q is a *measure*, not a probability distribution: summing over $\mathcal{T}$ gives $\sum_{\tau} q(\tau) = |\mathcal{A}|^T$, because q integrates out the environment but not the actions. It is exactly the “uniform over your own choices, honest about nature’s” prior. Nothing below requires q to be normalised.

Definition 5.3 (Relative entropy objective).

$$\mathcal{H}_q(p) \triangleq - \sum_{\tau \in \mathcal{T}} p(\tau) \log \frac{p(\tau)}{q(\tau)} = -\text{KL}(p \parallel q). \quad (15)$$

When P is deterministic and d_0 uniform, q is constant on feasible trajectories and $\mathcal{H}_q$ reduces to Shannon entropy up to an additive constant. That special case is the one usually presented; it is not the general case.

Two errors this definition prevents

(1) **The retrofit error.** If one maximises plain Shannon entropy, the stationary solution is $p(\tau) \propto e^{w^\top \Phi(\tau)}$ with *no dynamics factor*. Many expositions derive that, and then later assert without comment that “really” $p(\tau) \propto q(\tau) e^{w^\top \Phi(\tau)}$. These are solutions to two

different programs. Putting q in the objective from line one makes the derivation and the final formula agree.

(2) The continuum error. If one starts with $-\int p \log p d\tau$ over a continuous trajectory space, the objective is a *differential* entropy: it is not invariant under reparameterisation of τ and it is not bounded above. The relative entropy $-\text{KL}(p\|q)$ has neither defect. A further reason to begin discrete is that the pathology is invisible until you try to make it rigorous.

5.2 The primal program

$$\begin{aligned} \max_p \quad & \mathcal{H}_q(p) = - \sum_{\tau} p(\tau) \log \frac{p(\tau)}{q(\tau)} \\ \text{s.t.} \quad & \sum_{\tau} p(\tau) \Phi(\tau) = \mu_E \quad (\text{feature matching}) \\ & \sum_{\tau} p(\tau) = 1, \quad p(\tau) \geq 0 \quad (\text{normalisation}) \end{aligned} \quad (16)$$

The feasible set is the intersection of the simplex with d affine constraints, and $\mathcal{H}_q$ is strictly concave on it, so the solution—if the feasible set is non-empty—is unique. Note this uniqueness is over *distributions*, not over rewards; Section 3 still applies.

5.3 Variational derivation

Step 1: the Lagrangian. Introduce $w \in \mathbb{R}^d$ for feature matching and $c \in \mathbb{R}$ for normalisation:

$$\mathcal{L}(p, w, c) = - \sum_{\tau} p(\tau) \log \frac{p(\tau)}{q(\tau)} + w^\top \left(\sum_{\tau} p(\tau) \Phi(\tau) - \mu_E \right) + c \left(\sum_{\tau} p(\tau) - 1 \right). \quad (17)$$

Step 2: stationarity. Differentiating with respect to the scalar $p(\tau)$ for a fixed τ ,

$$\frac{\partial \mathcal{L}}{\partial p(\tau)} = - \log \frac{p(\tau)}{q(\tau)} - 1 + w^\top \Phi(\tau) + c = 0. \quad (18)$$

Step 3: solve.

$$p(\tau) = q(\tau) \exp(w^\top \Phi(\tau) + c - 1). \quad (19)$$

Step 4: normalise. Imposing $\sum_{\tau} p(\tau) = 1$ gives $e^{c-1} = 1/Z(w)$ with

Definition 5.4 (Partition function).

$$Z(w) \triangleq \sum_{\tau \in \mathcal{T}} q(\tau) e^{w^\top \Phi(\tau)}. \quad (20)$$

The maximum-entropy trajectory model

$$p_w(\tau) = \frac{1}{Z(w)} \underbrace{q(\tau)}_{\text{what nature does}} \underbrace{e^{w^\top \Phi(\tau)}}_{\text{what the agent wants}}, \quad Z(w) = \sum_{\tau} q(\tau) e^{w^\top \Phi(\tau)}. \quad (21)$$

Two trajectories with equal reward need *not* be equally likely: the one the environment resists is rarer. Exponentially better trajectories are exponentially more likely, but never certain—which is precisely the graceful handling of a sub-optimal expert that max-margin lacked.

5.4 The dual is maximum likelihood, and it is convex

Substituting (21) back into (17): since $\log p_w(\tau) - \log q(\tau) = w^\top \Phi(\tau) - \log Z(w)$, the entropy term becomes $-w^\top \sum_\tau p \Phi + \log Z(w)$, the normalisation term vanishes, and the feature term cancels the first piece, leaving

$$g(w) \triangleq \log Z(w) - w^\top \mu_E, \quad \text{to be minimised over } w. \quad (22)$$

Theorem 5.5 (Duality and maximum likelihood). *Minimising (22) is exactly maximising the average log-likelihood of the demonstrations under p_w :*

$$\begin{aligned} \frac{1}{m} \sum_{i=1}^m \log p_w(\tau^{(i)}) &= \frac{1}{m} \sum_{i=1}^m \left[\log q(\tau^{(i)}) + w^\top \Phi(\tau^{(i)}) - \log Z(w) \right] \\ &= \underbrace{w^\top \mu_E - \log Z(w)}_{=-g(w)} + \underbrace{\frac{1}{m} \sum_i \log q(\tau^{(i)})}_{\text{does not involve } w}. \end{aligned} \quad (23)$$

The dropped term is the log-likelihood of the observed transitions under the true dynamics; it is a constant of the optimisation, which is why it is usually invisible.

Theorem 5.6 (Convexity). *$w \mapsto \log Z(w)$ is convex, with*

$$\nabla_w \log Z(w) = \mathbb{E}_{p_w}[\Phi(\tau)], \quad \nabla_w^2 \log Z(w) = \text{Cov}_{p_w}[\Phi(\tau)] \succeq 0. \quad (24)$$

Hence g is convex; it is strictly convex, and the optimal w unique, precisely when the features Φ are not almost-surely confined to a proper affine subspace.

Proof. $\nabla_w Z = \sum_\tau q(\tau) e^{w^\top \Phi} \Phi$, so $\nabla_w \log Z = \sum_\tau p_w(\tau) \Phi(\tau)$. Differentiating once more gives $\mathbb{E}_{p_w}[\Phi \Phi^\top] - \mathbb{E}_{p_w}[\Phi] \mathbb{E}_{p_w}[\Phi]^\top$, a covariance matrix. $\square$

Corollary 5.7 (The learning rule).

$$\boxed{-\nabla_w g(w) = \mu_E - \mathbb{E}_{p_w}[\Phi(\tau)]} \quad (25)$$

Gradient ascent moves w so as to close the gap between what the expert did and what the current model predicts. At the optimum the two feature expectations coincide.

Remark 5.8 (A pleasing tension worth noticing). Theorem 5.6 says the reward *parameter* w has a unique optimum. Section 3 says the reward *function* is not identified. Both are true, and the resolution is that maximum entropy is a selection principle: it picks one point out of the equivalence class, and picks it uniquely. Uniqueness of an estimator is not identifiability of an estimand.

5.5 Computing Z : the whole difficulty, made concrete

Equation (25) looks like a two-line algorithm, and the entire remainder of these notes exists because of the innocuous term $\mathbb{E}_{p_w}[\Phi]$. Evaluating it requires $Z(w)$, and $Z(w)$ is a sum over $\mathcal{T}$.

Enumeration is hopeless but finite. With $|\mathcal{A}|$ actions and horizon T , $|\mathcal{T}|$ grows like $|\mathcal{A}|^T$ (times the branching of P). For a 5×5 gridworld with 4 actions and $T = 20$ that is $4^{20} \approx 10^{12}$ trajectories. Finite, and still impossible. The value of the discrete formulation is not that we enumerate; it is that Z is unambiguously *defined*, so we can look for structure.

The structure is that q factorises. Because $q(\tau) = d_0(s_0) \prod_t P(s_{t+1} | s_t, a_t)$ and $\Phi(\tau) = \sum_t \phi(s_t, a_t)$, the sum over trajectories telescopes into a backward recursion over states. Write $R = R_w$ for brevity.

Proposition 5.9 (Exact backward recursion for Z). Define $Z_T(s) \triangleq 1$ and, for $t = T-1, \dots, 0$,

$$Z_t(s) = \sum_{a \in \mathcal{A}} e^{R(s,a)} \sum_{s' \in \mathcal{S}} P(s' | s, a) Z_{t+1}(s'). \quad (26)$$

Then $Z(w) = \sum_s d_0(s) Z_0(s)$. Writing $V_t(s) = \log Z_t(s)$, this is

$$V_t(s) = \log \sum_a \exp \left(R(s, a) + \underbrace{\log \sum_{s'} P(s' | s, a) e^{V_{t+1}(s')}}_{=: Q_t(s, a)} \right). \quad (27)$$

The cost is $O(T|\mathcal{S}|^2|\mathcal{A}|)$ instead of $O(|\mathcal{A}|^{|\mathcal{S}|})$.

Proof. Let $Z_t(s)$ be the sum, over all suffixes $(a_t, s_{t+1}, \dots, s_T)$ started at $s_t = s$, of $\prod_{u \geq t} P(s_{u+1} | s_u, a_u) \exp(\sum_{u \geq t} R(s_u, a_u))$. Splitting off the first action and successor gives exactly (26); the base case is the empty suffix. Averaging over $s_0 \sim d_0$ gives $Z(w)$. $\square$

Answer: where does knowledge of P enter?

Look at (26). The kernel $P(s' | s, a)$ is consumed inside the inner sum, once per (s, a) per timestep. This is the **first** of exactly two places where MaxEnt IRL needs a model. The second appears in Section 5.8. When we ask, in Section 8, “what if P is unknown?”, these two sums are the entire content of the question.

5.6 A worked example you can check by hand

Example 5.10 (Two states, two actions, horizon two). Let $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{L, R\}$, $d_0 = \delta_{s_1}$, $T = 2$, and deterministic dynamics: L moves to s_1 , R moves to s_2 , from either state. Let $R(s_1, L) = 0$, $R(s_1, R) = 1$, $R(s_2, L) = 0$, $R(s_2, R) = 2$.

By enumeration. Four action sequences:

sequence	path	return	weight e^{return}
LL	$s_1 \rightarrow s_1 \rightarrow s_1$	$0 + 0 = 0$	1.0000
LR	$s_1 \rightarrow s_1 \rightarrow s_2$	$0 + 1 = 1$	2.7183
RL	$s_1 \rightarrow s_2 \rightarrow s_1$	$1 + 0 = 1$	2.7183
RR	$s_1 \rightarrow s_2 \rightarrow s_2$	$1 + 2 = 3$	20.0855
$Z =$			26.5221

By the backward recursion (27). With $V_2 \equiv 0$:

$$\begin{aligned} V_1(s_1) &= \log(e^0 + e^1) = 1.3133, & V_1(s_2) &= \log(e^0 + e^2) = 2.1269, \\ Q_0(s_1, L) &= 0 + V_1(s_1) = 1.3133, & Q_0(s_1, R) &= 1 + V_1(s_2) = 3.1269, \\ V_0(s_1) &= \log(e^{1.3133} + e^{3.1269}) = 3.2780, & Z &= e^{V_0(s_1)} = 26.522. \checkmark \end{aligned}$$

And the model’s first action probability agrees both ways:

$$p_w(a_0 = R) = e^{Q_0(s_1, R) - V_0(s_1)} = \frac{22.804}{26.522} = 0.8598 = \frac{e^1 + e^3}{Z} = \frac{22.804}{26.522}. \checkmark$$

Four trajectories collapsed into four (s, a) evaluations. With $T = 20$ the same collapse turns the $4^{20} \approx 10^{12}$ trajectories of the 5×5 gridworld into $25 \times 4 \times 20 = 2000$ state-action-time evaluations.

5.7 The stochastic-dynamics trap, and maximum *causal* entropy

Example 5.10 was deterministic, and that was not laziness. Look again at (27). The inner term

$$\log \sum_{s'} P(s' | s, a) e^{V_{t+1}(s')} \quad (28)$$

is a *soft maximum* over successor states, not an expectation. Since $\log \mathbb{E}[e^X] \geq \mathbb{E}[X]$ by Jensen, it systematically over-values actions with high variance in their outcomes.

The pathology is sharper than a bias in a formula. It is a statement about what the model believes.

Proposition 5.11 (The naive model secretly rewrites the dynamics). *Under p_w of (21), the conditional law of the successor state is*

$$p_w(s_{t+1} = s' | s_t = s, a_t = a) = \frac{P(s' | s, a) e^{V_{t+1}(s')}}{\sum_{s''} P(s'' | s, a) e^{V_{t+1}(s'')}} \neq P(s' | s, a) \quad \text{in general.} \quad (29)$$

Proof. Marginalise (21) over all suffixes after s_{t+1} using Proposition 5.9; the suffix sum contributes the factor $Z_{t+1}(s') = e^{V_{t+1}(s')}$. $\square$

What this means

The maximum-entropy trajectory model with stochastic dynamics is not a model of an agent choosing actions in the given MDP. It is a model of an agent who *also* tilts nature towards fortunate outcomes. Fitted to a real expert, it will explain lucky outcomes as intentional and infer rewards that reward luck. In a deterministic MDP the tilt is vacuous (P is a point mass), which is why the original presentation is correct as stated and why the problem is so easy to miss.

The remedy is to stop optimising over arbitrary trajectory distributions and optimise over *policies*, forcing the true dynamics to appear as an unchangeable factor.

Definition 5.12 (Causal entropy). For a (possibly time-dependent) policy π , let $p_\pi(\tau) = d_0(s_0) \prod_t \pi_t(a_t | s_t) P(s_{t+1} | s_t, a_t)$ and

$$\mathcal{H}^{\text{causal}}(\pi) \triangleq \mathbb{E}_{p_\pi} \left[\sum_{t=0}^{T-1} -\log \pi_t(a_t | s_t) \right]. \quad (30)$$

Only the agent's own choices are counted as uncertainty; randomness supplied by nature is not credited to the agent.

Theorem 5.13 (Maximum causal entropy IRL, Ziebart 2010 [5]). *The solution of*

$$\max_{\pi} \mathcal{H}^{\text{causal}}(\pi) \quad \text{s.t.} \quad \mathbb{E}_{p_\pi}[\Phi(\tau)] = \mu_E \quad (31)$$

is the soft-optimal policy for $R_w = w^\top \phi$ with w the multiplier, given by the backward recursion

$$Q_t(s, a) = R_w(s, a) + \sum_{s'} P(s' | s, a) V_{t+1}(s'), \quad (32)$$

$$V_t(s) = \log \sum_a \exp Q_t(s, a), \quad (33)$$

$$\pi_t(a | s) = \exp(Q_t(s, a) - V_t(s)), \quad (34)$$

with $V_T \equiv 0$. The dual objective remains convex and its gradient remains $\mu_E - \mathbb{E}_{p_\pi}[\Phi]$.

The one-symbol difference

Compare (27) with (32)–(33). Naive MaxEnt takes a *soft* maximum over both actions and successor states; maximum causal entropy takes a soft maximum over actions and a *plain expectation* over successor states:

$$\begin{aligned} \text{naive:} \quad & \log \sum_a \exp(R + \log \sum_{s'} P e^V) \\ \text{causal:} \quad & \log \sum_a \exp(R + \sum_{s'} P V) \end{aligned}$$

They coincide exactly when P is deterministic. Every modern method—soft Q -learning, soft actor–critic, GAIL’s entropy term, AIRL—uses the causal version. When a paper says “MaxEnt IRL,” it almost always means this.

5.8 The forward pass: from a policy to expected features

The backward pass produced a policy. We still need $\mathbb{E}_{p_\pi}[\Phi]$, which requires knowing *where* that policy goes. That is a forward sweep.

Proposition 5.14 (Forward state-visitation recursion). *Let $d_t(s) = \Pr_{p_\pi}(s_t = s)$. Then d_0 is the initial distribution and*

$$d_{t+1}(s') = \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} d_t(s) \pi_t(a | s) P(s' | s, a), \quad (35)$$

whence

$$\mathbb{E}_{p_\pi}[\Phi] = \sum_{t=0}^{T-1} \sum_{s,a} d_t(s) \pi_t(a | s) \phi(s, a). \quad (36)$$

Answer: where does knowledge of P enter? (second and last place)

Equation (35). Together with (32), these are the *only* two consumers of P in exact MaxEnt IRL. A backward pass computes *what the agent would do*; a forward pass computes *where doing it takes them*. Both need the kernel; neither needs anything else.

5.9 Occupancy measures: the same object, better named

Equation (36) is a weighted average of ϕ over state–action pairs. Naming that weighting will pay for itself many times over.

Definition 5.15 (Discounted occupancy measure).

$$\rho_\pi(s, a) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \Pr_\pi(s_t = s, a_t = a), \quad (37)$$

a probability distribution over $\mathcal{S} \times \mathcal{A}$. Then

$$\mu(\pi) = \frac{1}{1 - \gamma} \sum_{s,a} \rho_\pi(s, a) \phi(s, a), \quad J_R(\pi) = \frac{1}{1 - \gamma} \sum_{s,a} \rho_\pi(s, a) R(s, a). \quad (38)$$

Remark 5.16 (Occupancy is not action frequency). ρ_π is a distribution over *pairs*. “Brake” in front of a pedestrian and “brake” on an empty motorway are different events because their states differ. MaxEnt IRL never merely counts action labels.

Equation (38) says return is a *linear functional* of occupancy. We will build all of Part V on that sentence.

5.10 The complete exact algorithm

Because the objective is convex (Theorem 5.6), any diminishing-step gradient scheme converges to the global optimum in w . The per-iteration cost is $O(T|\mathcal{S}|^2|\mathcal{A}|)$ for each pass—linear in the horizon, quadratic in the number of states. That quadratic is the thing that will not survive contact with a real problem.

Algorithm 2 Exact Maximum Causal Entropy IRL (finite $\mathcal{S}, \mathcal{A}$, known P)

Require: dynamics $\mathcal{M}$ with known P , features ϕ , expert data $\mathcal{D}_E$, step size η

```

1:  $\mu_E \leftarrow \frac{1}{m} \sum_i \Phi(\tau^{(i)})$ 
2: initialise  $w \leftarrow 0$ 
3: for  $k = 1, 2, \dots$  until  $\|\mu_E - \hat{\mu}\|$  small do
4:    $R_w(s, a) \leftarrow w^\top \phi(s, a)$  for all  $(s, a)$ 
5:   backward pass:  $V_T \equiv 0$ ; for  $t = T - 1, \dots, 0$  compute  $Q_t, V_t, \pi_t$  by (32)–(34)  $\triangleright$  uses  $P$ 
6:   forward pass:  $d_0 \leftarrow d_0$ ; for  $t = 0, \dots, T - 1$  propagate  $d_{t+1}$  by (35)  $\triangleright$  uses  $P$ 
7:    $\hat{\mu} \leftarrow \sum_{t,s,a} d_t(s) \pi_t(a | s) \phi(s, a)$ 
8:    $w \leftarrow w + \eta (\mu_E - \hat{\mu})$   $\triangleright$  gradient (25)
9: end for
10: return  $w$ , and the soft-optimal policy  $\pi$ 

```

Ledger: Exact MaxEnt / MaxCausalEnt IRL

Assumed (Coordinate 1). Expert is *Boltzmann-rational*: exponentially prefers better trajectories rather than always choosing the best. Reward linear in fixed ϕ . Among all models matching μ_E , take the maximum-causal-entropy one.

Approximated (Coordinate 2). Nothing. Z is exact via a backward pass; $\mathbb{E}[\Phi]$ is exact via a forward pass. Requires: finite $\mathcal{S}, \mathcal{A}$, and the kernel P in closed form.

Delivers. A unique w (convex problem), a stochastic policy that degrades gracefully with expert noise, and a likelihood—so the model can be compared, regularised and given priors, none of which max-margin allowed.

Breaks when. $|\mathcal{S}|$ is large or continuous, the horizon is infinite, or P is unknown. These are Sections 6, 7 and 8.

Part IV

Three Exits from the Safe Corner

Algorithm 2 is a complete, correct, globally convergent IRL method. It is also unusable on any problem anyone actually cares about, for three separate reasons. Each reason is one of the assumptions we made on entering Part III, and each is a question every reader of the IRL literature ends up asking. We take them one at a time, and in each case we ask the same two things: *what survives, and what must be replaced?*

6 Exit A: What Changes When the Problem Is Not Finite?

“Not finite” hides three separate difficulties: an unbounded horizon, an unbounded number of states, and a continuum of actions. They fail differently.

6.1 Infinite horizon: discounting, and why trajectory entropy dies

The finite-horizon derivation of Section 5 does not survive $T \rightarrow \infty$ as written, and the reason is not the return but the *entropy*. The causal entropy $\mathbb{E}[\sum_{t < T} -\log \pi(a_t | s_t)]$ grows linearly in T for any non-deterministic policy, so the objective diverges. The repair is the same discounting we already apply to reward.

Definition 6.1 (Discounted causal entropy and the soft objective).

$$\mathcal{H}_\gamma(\pi) \triangleq \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t (-\log \pi(a_t | s_t)) \right], \quad J_R^\beta(\pi) \triangleq J_R(\pi) + \beta \mathcal{H}_\gamma(\pi). \quad (39)$$

The temperature $\beta > 0$ is now explicit; Section 5 used $\beta = 1$.

Definition 6.2 (Soft Bellman operator).

$$(\mathcal{B}_{\text{soft}} Q)(s, a) \triangleq R(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} \left[\beta \log \sum_{a'} \exp(Q(s', a')/\beta) \right]. \quad (40)$$

Lemma 6.3 (Log-sum-exp is a non-expansion). *For $x, y \in \mathbb{R}^{|\mathcal{A}|}$ and $\beta > 0$, $|\beta \log \sum_a e^{x_a/\beta} - \beta \log \sum_a e^{y_a/\beta}| \leq \|x - y\|_\infty$.*

Proof. Let $\delta = \|x - y\|_\infty$. Then $x_a \leq y_a + \delta$ for all a , so $\sum_a e^{x_a/\beta} \leq e^{\delta/\beta} \sum_a e^{y_a/\beta}$; taking $\beta \log$ gives $\beta \log \sum e^{x_a/\beta} \leq \beta \log \sum e^{y_a/\beta} + \delta$. Exchanging x and y gives the other inequality. $\square$

Theorem 6.4 (Soft value iteration converges). *$\mathcal{B}_{\text{soft}}$ is a γ -contraction in $\|\cdot\|_\infty$, so it has a unique fixed point Q_{soft} , soft value iteration converges geometrically, and the soft-optimal policy is $\pi_{\text{soft}}(a | s) \propto \exp(Q_{\text{soft}}(s, a)/\beta)$, which maximises (39).*

Proof. Apply Lemma 6.3 pointwise inside the expectation in (40) and use $\sum_{s'} P = 1$. $\square$

Remark 6.5 (The temperature interpolates). Since

$$\max_a Q(s, a) \leq \beta \log \sum_a e^{Q(s, a)/\beta} \leq \max_a Q(s, a) + \beta \log |\mathcal{A}|,$$

the soft operator is within $\beta \log |\mathcal{A}|$ of $\mathcal{B}^*$ everywhere. As $\beta \rightarrow 0$ we recover hard optimality and max-margin’s behaviour model; as $\beta \rightarrow \infty$ the policy becomes uniform and the demonstrations are treated as uninformative. *The temperature is a modelling choice on Coordinate 1*: it says how rational you think the expert is. Fitting β from data is possible but is entangled with the scale of R , since (R, β) and $(cR, c\beta)$ give the same policy—one more instance of the identifiability theme.

What survived Exit A, part 1

The whole structure of Section 5 survives an infinite horizon, provided we use *discounted causal* entropy and a stationary soft Bellman fixed point in place of a finite backward sweep. What we lose is only the interpretation of $\log Z$ as a partition function of a finite trajectory set; what replaces it is a soft value function. Nothing statistical changes.

6.2 Large or continuous state spaces

Here the loss is real. Both passes of Algorithm 2 are tabular sweeps costing $O(|\mathcal{S}|^2|\mathcal{A}|)$. Three things break at once.

1. **The tables do not fit.** V , π and d_t are arrays indexed by $\mathcal{S}$. For continuous $\mathcal{S}$ they are functions and must be approximated—by linear architectures, or in practice by neural networks. This reintroduces every hazard of the deadly triad into the inner loop of IRL.
2. **Discretisation does not save you.** An ε -net of $[0, 1]^n$ has ε^{-n} cells. A ten-dimensional state at resolution 0.1 is 10^{10} cells, and the sweep is quadratic in that.
3. **Sums become integrals and the reference measure becomes load-bearing.** In (20), $\sum_{\tau} q(\tau)(\cdot)$ becomes $\int(\cdot) dq(\tau)$. This is precisely where the relative-entropy formulation of Definition 5.1 pays off: $-\text{KL}(p||q)$ is well-defined and reparameterisation-invariant on a continuous space, whereas $-\int p \log p$ is neither. Had we begun with the continuous differential entropy, the objective would have changed value under a smooth change of coordinates on $\mathcal{T}$ —an absurdity for a principle that is supposed to encode ignorance.

6.3 Continuous actions

The soft-max over actions, $\beta \log \sum_a \exp(Q/\beta)$, becomes $\beta \log \int_{\mathcal{A}} \exp(Q(s, a)/\beta) da$: an integral with no closed form for a general Q . The standard resolution is variational: represent the policy explicitly and fit it to the Boltzmann target,

$$\pi \leftarrow \arg \min_{\pi' \in \Pi_{\theta}} \text{KL} \left(\pi'(\cdot | s) \parallel \frac{1}{Z(s)} \exp(Q(s, \cdot)/\beta) \right), \quad (41)$$

which is exactly the policy-improvement step of soft actor-critic [13], and an instance of the broader control-as-inference picture in which entropy-regularised RL is exact probabilistic inference in a graphical model [14]. Note the structure: an intractable normaliser over actions is replaced by a learned sampler. We are about to meet the same move one level up, where an intractable normaliser over *trajectories* is replaced by a learned policy.

6.4 Summary of Exit A

Object	Finite case	Infinite / continuous case
Entropy objective	causal entropy over T steps	discounted causal entropy; relative to q
Backward pass	exact sweep, T steps	soft Bellman fixed point (contraction survives)
Soft-max over actions	$\log \sum_a$	intractable integral; variational policy (SAC)
Forward pass / ρ_{π}	exact array recursion	Monte Carlo rollouts or density models
Partition function Z	finite sum, exact via DP	intractable; estimated or bypassed
Convexity of the dual in w	holds	holds <i>if</i> the reward stays linear in w

The last row deserves emphasis: leaving the safe corner along Exit A alone does *not* cost us convexity. Convexity is lost at Exit B.

7 Exit B: Must the Features Be Hand-Crafted?

Short answer: no, and modern methods do not. But the question hides a more useful one—*what was the fixed ϕ doing for us?*—and the honest answer is: quite a lot.

7.1 What a fixed linear class buys

1. **A sufficient statistic.** By Proposition 3.4, a policy is summarised for all reward-evaluation purposes by $\mu(\pi) \in \mathbb{R}^d$. An infinite-dimensional object collapses to d numbers.
2. **Convexity.** Theorem 5.6 needs R linear in the parameter. This is what makes MaxEnt IRL a well-behaved convex estimation problem.
3. **A class-wide guarantee.** Lemma 4.2 certifies performance for *every* reward in the class at once, which is much stronger than certifying it for one estimate.
4. **Interpretability.** “The expert weights collision risk 8 times lane deviation” is a sentence one can act on.

7.2 What it costs

Misspecification is invisible from inside

Every guarantee above is quantified over $w \in \mathbb{R}^d$ with ϕ fixed. If the true reward is not in $\text{span}\{\phi_1, \dots, \phi_d\}$, matching μ_E implies *nothing* about J_{R^*} . Worse, the fit will look fine: the residual lives in a direction the model cannot see. Classical apprenticeship-learning results on driving worked because a human had already encoded the task into features such as “distance to nearest car” and “off-road indicator”. The domain knowledge was in ϕ , not the algorithm.

7.3 The ambiguity that appears when ϕ is learned

Suppose we learn ϕ_η and w jointly. For any invertible $A \in \mathbb{R}^{d \times d}$,

$$w^\top \phi(s, a) = (A^{-\top} w)^\top (A\phi(s, a)), \quad (42)$$

so (w, ϕ) and $(A^{-\top} w, A\phi)$ are indistinguishable. Only the *scalar field* R and the *span* of the features are meaningful; individual coordinates of w are not. Statements like “the model learned that feature 3 matters” are therefore meaningless without a convention pinning down A . This is a Coordinate-1 ambiguity *added* by learning features, on top of the reward ambiguity we already had.

7.4 The good news: the learning rule does not change

One might fear that abandoning linearity forces a new derivation. It does not.

Proposition 7.1 (MaxEnt gradient for an arbitrary reward class). *Let R_θ be any differentiable reward parameterisation, and let $R_\theta(\tau) = \sum_t \gamma^t R_\theta(s_t, a_t)$. With $p_\theta(\tau) \propto q(\tau) e^{R_\theta(\tau)}$, the log-likelihood $L(\theta) = \frac{1}{m} \sum_i R_\theta(\tau^{(i)}) - \log Z(\theta)$ has gradient*

$$\nabla_\theta L(\theta) = \underbrace{\mathbb{E}_{\tau \sim \mathcal{D}_E} [\nabla_\theta R_\theta(\tau)]}_{\text{expert}} - \underbrace{\mathbb{E}_{\tau \sim p_\theta} [\nabla_\theta R_\theta(\tau)]}_{\text{current model}}. \quad (43)$$

Proof. $\nabla_{\theta} \log Z(\theta) = \frac{1}{Z} \sum_{\tau} q(\tau) e^{R_{\theta}(\tau)} \nabla_{\theta} R_{\theta}(\tau) = \mathbb{E}_{p_{\theta}}[\nabla_{\theta} R_{\theta}(\tau)]$. $\square$

Equation (43) is (25) with ϕ replaced by $\nabla_{\theta} R_{\theta}$. In the linear case $\nabla_w R_w(\tau) = \Phi(\tau)$ and the two coincide exactly. *Feature matching was never the principle; matching the sufficient statistics of the reward class was.* This observation is what makes deep MaxEnt IRL [12] and guided cost learning [11] possible with no new theory.

7.5 What is lost: convexity, and the need for inductive bias

With R_{θ} nonlinear, L is no longer concave. We get local optima, sensitivity to initialisation, and—most importantly—capacity to overfit. A sufficiently expressive R_{θ} can memorise the demonstrations: assign huge reward exactly on the observed state-action pairs and zero elsewhere. That reward perfectly explains the data and is useless.

Useful counter-pressures, all of which are Coordinate-1 commitments:

- **Architectural restriction.** Most powerfully, making R depend on the state only, $R_{\theta}(s)$. We will see in Section 11 that this is not a convenience but a necessary condition for a transfer guarantee.
- **Smoothness and norm penalties** on R_{θ} ; gradient penalties on a discriminator.
- **Bottlenecks**, including explicit information bottlenecks on the reward representation.
- **Multi-task structure:** several tasks sharing one representation with different weights, which restores identifiability of the shared part.
- **Priors**, i.e. Bayesian IRL [15], which makes the selection principle explicit rather than implicit.

Answer to Exit B

No, ϕ need not be hand-crafted, and the maximum-entropy learning rule (43) is unchanged when it is not. What changes is that the estimation problem stops being convex and starts requiring regularisation, and that a new reparameterisation ambiguity is introduced. The choice of reward class does not disappear when features are learned—it becomes the choice of *architecture*. In Part V it becomes the choice of *discriminator*, which is why we insist that a discriminator architecture is a reward class in disguise.

8 Exit C: What If We Do Not Know P ?

This is the question that most often derails a first reading, because two entirely different estimation problems are usually collapsed into one. Separate them first.

Two statistical jobs, frequently confused

Job 1 — estimate the expert’s statistics ρ_E (or μ_E). Data: the demonstrations. Needs no model and no interaction: an average over $\mathcal{D}_E$. Its weakness is *support*—it says nothing outside the states the expert visited.

Job 2 — estimate the *current candidate policy’s* statistics ρ_{π_w} . This is not in the data at all: π_w changes every gradient step. It requires either a model of P (to compute) or the ability to interact with the environment (to sample). This is the job that makes IRL expensive, and the only one that needs P .

Recall from Sections 5.5 and 5.8 that P is consumed in exactly two places, both belonging to Job 2: the expectation over successors in the backward pass (32), and the propagation of visitation in the forward pass (35). Everything below is about replacing those two sums.

8.1 Four regimes

What you have	What to do about Job 2	Price
P in closed form, finite $\mathcal{S}$	Exact backward + forward passes (Algorithm 2)	$O(T \mathcal{S} ^2 \mathcal{A})$ per gradient step
A simulator you can re-set	Roll out π_w ; estimate $\mathbb{E}[\Phi]$ by Monte Carlo. You never need to <i>write down</i> P	Variance; many rollouts per gradient step
Only the real environment	Fit $\hat{P}$ and plan, or estimate occupancies directly from on-policy data	Exploration cost; model bias; safety risk
Only a fixed offline dataset	Job 2 is not identifiable off-support. Requires pessimism, conservative penalties, or restricting the policy class	Genuinely hard; an active research area

Remark 8.1 (A simulator is not a model, and that is the point). The second row is the one that dissolves most of the confusion. “Model-free” does not mean “dynamics-free”. It means you never need P as an object; you need only the ability to draw $s' \sim P(\cdot | s, a)$. Almost every modern IRL method lives in this row.

8.2 Sampling the gradient

The Monte Carlo replacement is direct. Draw n rollouts $\tau_1, \dots, \tau_n$ from the current soft policy and set

$$\widehat{\mathbb{E}_{p_w}[\Phi]} = \frac{1}{n} \sum_{j=1}^n \Phi(\tau_j), \quad \widehat{\nabla_w L} = \mu_E - \frac{1}{n} \sum_{j=1}^n \Phi(\tau_j), \quad (44)$$

which is unbiased and consistent. The catch is that the current soft policy must itself be obtained by an inner forward-RL solve at every outer step—extremely wasteful, since w changes only slightly.

Guided cost learning [11] removes the waste with importance sampling: maintain a sampler π_{samp} , reuse its rollouts, and correct with weights

$$\omega_j \propto \frac{e^{R_\theta(\tau_j)}}{\pi_{\text{samp}}(\tau_j)}, \quad \widehat{\nabla_\theta L} = \mathbb{E}_{\mathcal{D}_E}[\nabla_\theta R_\theta] - \sum_j \frac{\omega_j}{\sum_{j'} \omega_{j'}} \nabla_\theta R_\theta(\tau_j), \quad (45)$$

interleaving improvement of π_{samp} towards the soft optimum. Note the dynamics terms cancel in the ratio—one never needs P —but the weights degenerate when π_{samp} drifts from p_θ , which is the practical failure mode.

8.3 Support, exploration, and model error

- **Support.** A transition never attempted cannot be estimated. If ρ_E has support the learner never visits (or vice versa), the comparison driving (25) is being made on incomparable sets. This is the deep reason imitation degrades off the expert’s distribution, and the origin of compounding error in behavioural cloning [23].
- **Model error compounds.** If one fits $\hat{P}$ and plans in it, errors accumulate over the effective horizon: a per-step total-variation error ε produces occupancy error of order $\varepsilon/(1 - \gamma)$ and hence a *biased* reward gradient. Bias does not average away over gradient steps the way variance does.

- **Interaction may be unacceptable.** Exploration in surgery, aviation or finance has costs that no sample-complexity bound captures.

A practical decision rule

Demand exact transition probabilities only when the algorithm asks for exact dynamic-programming expectations. If you can *sample* transitions, every such expectation can be estimated without ever writing P down. What you pay is sample complexity, variance, a coverage requirement, and possibly interaction risk. What you do *not* pay is any change to the IRL objective itself: Coordinate 1 is untouched by Exit C.

8.4 The pivot: from estimating an expectation to estimating a ratio

Here is the idea that generates the rest of these notes.

The gradient (25) asks us to compare two expectations of a *known* function ϕ under two *unknown* distributions, ρ_E and ρ_π . We have samples from both: the demonstrations give samples of ρ_E , and rollouts give samples of ρ_π . So the real task is: *given samples from two distributions, quantify and reduce their discrepancy.*

Machine learning has an extremely well-developed tool for exactly that task, and it is not integration. It is **binary classification**. A classifier trained to distinguish samples of ρ_E from samples of ρ_π recovers, at its optimum, the density ratio between them—and the density ratio is precisely the discrepancy signal we wanted. No partition function, no transition kernel, no enumeration.

That substitution is GAIL. We now make it precise.

Part V

Adversarial Methods

9 Occupancy Measures: the Right Change of Variables

Sections 4 and 5 both optimise over policies, and both objectives turned out to depend on the policy only through averages of the form $\sum_{s,a} \rho_\pi(s,a)(\cdot)$. That is a strong hint that ρ , not π , is the natural variable. Changing to it converts IRL into a convex-duality problem and makes GAIL a theorem rather than an analogy.

Definition 9.1 (The occupancy polytope). Let

$$\mathcal{D} \triangleq \left\{ \rho \in \mathbb{R}_{\geq 0}^{S \times A} : \sum_a \rho(s,a) = (1-\gamma)d_0(s) + \gamma \sum_{s',a'} P(s | s', a') \rho(s', a') \quad \forall s \right\}, \quad (46)$$

the set of non-negative measures satisfying the *Bellman flow* constraints.

Theorem 9.2 (Policies and occupancies are the same thing). *The map $\pi \mapsto \rho_\pi$ defined by (37) is a bijection from stationary policies (identified when they agree on every state they visit) onto $\mathcal{D}$, with inverse*

$$\pi_\rho(a | s) = \frac{\rho(s,a)}{\sum_{a'} \rho(s,a')}. \quad (47)$$

$\mathcal{D}$ is a compact convex polytope.

This is classical [7, 8]. Three consequences do all the work:

1. **Return is linear in ρ :** $J_R(\pi) = \frac{1}{1-\gamma} \langle R, \rho_\pi \rangle$ by (38). Forward RL is therefore a linear program over $\mathcal{D}$.
2. **Matching occupancy is matching behaviour:** by Theorem 9.2, $\rho_\pi = \rho_E$ if and only if $\pi = \pi_E$ on every state the expert visits. Imitation is exactly a distribution-matching problem, with no reward mentioned.
3. **Mixtures are legal:** $\mathcal{D}$ is convex, so $\rho_{\bar{\pi}} = \sum_i \lambda_i \rho_{\pi_i}$ is realisable—which is what justified the mixed policy in Algorithm 1.

Finally, discounted causal entropy is a concave function of ρ :

$$\bar{\mathcal{H}}(\rho) = - \sum_{s,a} \rho(s,a) \log \frac{\rho(s,a)}{\sum_{a'} \rho(s,a')}, \quad (48)$$

so “entropy-regularised RL” is a concave maximisation over a polytope.

10 GAIL: Imitation as Density-Ratio Estimation

10.1 IRL with an explicit reward regulariser

Write IRL abstractly, following Ho and Ermon’s variational treatment [9]. A reward is good if the expert scores well under it and the best achievable policy does not score much better—i.e. if the reward *explains the gap*. Add a convex regulariser ψ on rewards to keep the problem bounded:

$$\text{IRL}_\psi(\pi_E) = \arg \max_R -\psi(R) + \left(\min_{\pi} -\beta \bar{\mathcal{H}}(\rho_\pi) - \langle R, \rho_\pi \rangle \right) + \langle R, \rho_E \rangle. \quad (49)$$

Without ψ this is exactly the ill-posed problem of Section 3; ψ is the selection principle, written down explicitly for once.

Theorem 10.1 (Ho & Ermon 2016). *Running RL on the reward returned by (49) yields*

$$\text{RL} \circ \text{IRL}_\psi(\pi_E) = \arg \min_{\pi} -\beta \bar{\mathcal{H}}(\rho_\pi) + \psi^*(\rho_\pi - \rho_E), \quad (50)$$

where ψ^* is the convex conjugate of ψ .

Read Theorem 10.1 slowly

It says that the whole IRL-then-RL pipeline is *equivalent* to a single occupancy-matching problem, in which the choice of reward regulariser ψ becomes the choice of *divergence* ψ^* used to compare ρ_π with ρ_E . The intermediate reward is a computational device that can be eliminated. Everything below is a choice of ψ .

10.2 Two choices of ψ

(i) A constant on a linear class. Let ψ be 0 if $R = w^\top \phi$ with $\|w\|_2 \leq 1$ and $+\infty$ otherwise. Then ψ^* is the support function of that set, and (50) becomes

$$\min_{\pi} \max_{\|w\|_2 \leq 1} w^\top (\mu(\pi) - \mu_E) = \min_{\pi} \|\mu(\pi) - \mu_E\|_2, \quad (51)$$

which is apprenticeship learning (Section 4) recovered exactly. Its weakness is now visible as a property of ψ^* : it is a hard constraint, forcing exact feature matching within a small class, and it treats any ρ_π outside that class as equally bad.

(ii) A smooth penalty on the expert's own reward values. Ho and Ermon choose $\psi_{\text{GA}}(R) = \mathbb{E}_{\rho_E}[g(R(s, a))]$ for a particular convex g , whose conjugate is a Jensen–Shannon divergence:

$$\psi_{\text{GA}}^*(\rho_\pi - \rho_E) = \max_{D: \mathcal{S} \times \mathcal{A} \rightarrow (0,1)} \mathbb{E}_{\rho_E}[\log D(s, a)] + \mathbb{E}_{\rho_\pi}[\log(1 - D(s, a))]. \quad (52)$$

This regulariser penalises rewards that are large on expert pairs, is defined for *all* rewards rather than a fixed span, and—crucially—its conjugate is computable from samples alone.

10.3 The GAIL objective

GAIL (Ho & Ermon, 2016)

$$\min_{\pi} \max_D \mathbb{E}_{(s,a) \sim \rho_E}[\log D(s, a)] + \mathbb{E}_{(s,a) \sim \rho_\pi}[\log(1 - D(s, a))] - \beta \bar{\mathcal{H}}(\rho_\pi). \quad (53)$$

Convention, fixed for the rest of these notes: $D(s, a)$ is the estimated probability that the pair came **from the expert**. (Ho and Ermon’s paper uses the opposite labelling; the objectives are related by $D \mapsto 1 - D$. Mixing the two conventions is the single most common source of sign confusion in this literature.)

Proposition 10.2 (The discriminator estimates a density ratio). *For fixed π , the inner maximum of (53) is attained at*

$$D^*(s, a) = \frac{\rho_E(s, a)}{\rho_E(s, a) + \rho_\pi(s, a)}, \quad (54)$$

and the optimal value equals $2 \text{JS}(\rho_E \| \rho_\pi) - \log 4$. Hence GAIL is

$$\min_{\pi} 2 \text{JS}(\rho_E \| \rho_\pi) - \beta \bar{\mathcal{H}}(\rho_\pi). \quad (55)$$

Proof. Pointwise in (s, a) , maximise $\rho_E \log D + \rho_\pi \log(1 - D)$ over $D \in (0, 1)$; the stationary point is D^* . Substituting gives the Jensen–Shannon expression. $\square$

10.4 Turning the discriminator into a reward

The policy player minimises $\mathbb{E}_{\rho_\pi}[\log(1 - D)]$, i.e. it maximises the return of the surrogate reward

$$\hat{r}(s, a) = -\log(1 - D(s, a)) \in (0, \infty). \quad (56)$$

Any policy-gradient method (TRPO, PPO) can then be run on $\hat{r}$.

Surrogate rewards are not interchangeable

Three surrogates appear in practice and they are *different functions*:

surrogate	range	implicit bias
$-\log(1 - D)$	$(0, +\infty)$	always positive: encourages <i>surviving</i> longer
$\log D$	$(-\infty, 0)$	always negative: encourages <i>terminating</i> sooner
$\log D - \log(1 - D)$	$(-\infty, \infty)$	the true log-odds; unbiased in sign

On tasks with variable episode length these choices change the learned behaviour markedly, independently of anything about imitation. A text that says “use the log-odds” and an equation that uses $-\log(1 - D)$ are not describing the same algorithm. AIRL uses the log-odds form, for reasons that will be structural rather than cosmetic.

Algorithm 3 GAIL (Ho & Ermon, 2016)

Require: expert data $\mathcal{D}_E$, environment sampler (no P needed), initial π_θ , discriminator D_φ

- 1: **for** $k = 1, 2, \dots$ **do**
- 2: collect rollouts $\mathcal{B} \sim \pi_\theta$ by interacting with the environment
- 3: **discriminator step:** ascend $\nabla_\varphi[\mathbb{E}_{\mathcal{D}_E} \log D_\varphi + \mathbb{E}_{\mathcal{B}} \log(1 - D_\varphi)]$
- 4: **policy step:** run one TRPO/PPO update on reward $\hat{r} = -\log(1 - D_\varphi)$ with entropy bonus β
- 5: **end for**
- 6: **return** π_θ

10.5 What GAIL does not give you

At convergence, the discriminator contains no reward information

If imitation succeeds then $\rho_\pi = \rho_E$, so $D^* \equiv \frac{1}{2}$ and the surrogate reward (56) is the *constant* $\log 2$. A constant reward makes every policy optimal (Section 3). There is therefore literally nothing to extract at the end of a successful GAIL run.

Any reward-like signal GAIL exhibits exists only *during* training, as a by-product of the current mismatch between learner and expert. It is a function of the optimisation trajectory, not of the task. This—not a vague appeal to “shaping terms”—is the precise reason GAIL does not transfer.

Ledger: GAIL

Assumed (Coordinate 1). Behaviour model: entropy-regularised optimality. Selection principle: the ψ_{GA} regulariser, equivalently “match occupancy in Jensen–Shannon divergence”. Reward class: whatever the discriminator architecture can express.

Approximated (Coordinate 2). Everything, and cheaply. Z is never formed; P is never written down; both occupancy measures are represented only by samples. Requires *online* interaction with the environment, and inherits the instability of minimax optimi-

sation.

Delivers. An imitating policy, often excellent, in continuous control from raw observations. **Not** a reward.

Therefore. Use GAIL when you want behaviour in the same MDP. If you want an objective you can re-optimize elsewhere, read on.

11 AIRL: Recovering Something That Transfers

AIRL [10] keeps GAIL’s sampling machinery and repairs its Coordinate-1 deficit by *constraining the discriminator* so that a transferable reward is forced to appear inside it.

11.1 Where the discriminator’s form comes from

It is not an arbitrary architecture. Suppose the “expert” distribution is the maximum-entropy model $p_\theta(\tau) \propto q(\tau)e^{\sum_t f_\theta(s_t, a_t)}$ of Section 5, and the “fake” samples come from the current policy π . Per transition, the Bayes-optimal discriminator between the two is

$$D_\theta(s, a) = \frac{(\text{expert density})}{(\text{expert density}) + (\text{policy density})} = \frac{\exp f_\theta(s, a)}{\exp f_\theta(s, a) + \pi(a | s)}, \quad (57)$$

because the dynamics factors q are common to both and cancel. Two facts follow at once.

Proposition 11.1 (The AIRL reward is the logit minus a log-policy). *With D as in (57), the log-odds reward is*

$$\hat{r}(s, a) = \log D_\theta(s, a) - \log(1 - D_\theta(s, a)) = f_\theta(s, a) - \log \pi(a | s). \quad (58)$$

Remark 11.2 (The $-\log \pi$ term is not noise). Maximising $\mathbb{E}_\pi[\sum_t \gamma^t (f - \log \pi)]$ is exactly maximising $J_f(\pi) + \mathcal{H}_\gamma(\pi)$: the entropy bonus of (39) appears automatically. The adversarial game is therefore performing entropy-regularised RL on f whether or not anyone adds an entropy term by hand. At the optimum f_θ recovers the *advantage* $A_{\text{soft}}(s, a) = Q_{\text{soft}} - V_{\text{soft}}$ of the expert’s soft policy—which is the crux of what follows, because an advantage is a reward plus shaping, not a reward.

11.2 The structural constraint

AIRL restricts f to a shaped form:

$$f_{\theta, \omega}(s, a, s') = \underbrace{g_\theta(s)}_{\text{task reward}} + \underbrace{\gamma h_\omega(s') - h_\omega(s)}_{\text{potential-based shaping}}. \quad (59)$$

g_θ must depend on the state only

This is the load-bearing assumption of AIRL and it is very often mis-stated as $g_\theta(s, a)$. Here is why it must be $g_\theta(s)$.

For fixed dynamics P , the rewards inducing the same soft-optimal policy are exactly those of the form

$$r'(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)}[\Psi(s')] - \Psi(s) \quad (60)$$

for some potential Ψ . Note the shaping term $\gamma \mathbb{E}_{s' | s, a}[\Psi(s')] - \Psi(s)$ **depends on** a , and **depends on** P . So a state–action reward is identified only up to a nuisance term that (i) cannot be distinguished from the reward by any amount of behavioural data, and (ii) *changes when the dynamics change*. That is precisely a failure of transfer.

If instead g is a function of s alone, the only member of the family (60) that is state-only is $\Psi \equiv \text{const}$ (under the condition below). The action-dependent freedom is squeezed entirely into h_ω , and g_θ is pinned down up to an additive constant.

Definition 11.3 (Decomposability condition). Call two states *one-step linked* if some state has positive-probability transitions to both. The dynamics are *decomposable* if the transitive closure of this relation connects all of $\mathcal{S}$. Equivalently: any two functions $a(s), b(s)$ with $a(s) + b(s')$ constant along all feasible transitions must themselves be constant.

Theorem 11.4 (Disentanglement, Fu et al. 2018). *Suppose the dynamics are decomposable, the discriminator is trained to optimality with f constrained as in (59), and the true expert is soft-optimal for a state-only reward $r^*(s)$. Then at the global optimum there are constants c_1, c_2 with*

$$g_\theta(s) = r^*(s) + c_1, \quad h_\omega(s) = V_{\text{soft}}^*(s) + c_2. \quad (61)$$

In particular g_θ is invariant to changes in the transition dynamics, so re-optimising g_θ in a different MDP with the same reward structure recovers appropriate behaviour there.

Remark 11.5 (What the theorem does and does not say). It says: under a structural condition on the dynamics, a state-only reward assumption, and exact optimisation, the recovered g equals the ground-truth reward up to a constant. It does *not* say the reward is uniquely recoverable in general; the state-only restriction is doing the identification, and it is an assumption about the world, not a fact derived from data. If the true reward genuinely depends on actions (an energy cost, say), AIRL will misattribute that dependence into h_ω and the recovered g will be wrong. Section 3 has not been repealed; it has been paid off with an extra assumption, and the notes' whole point is that this is always the trade.

Algorithm 4 AIRL (Fu, Luo & Levine, 2018)

Require: expert data $\mathcal{D}_E$, environment sampler, policy π_θ , networks g_θ, h_ω

- 1: **for** $k = 1, 2, \dots$ **do**
 - 2: collect rollouts $\mathcal{B} \sim \pi_\theta$
 - 3: form $f_{\theta, \omega}(s, a, s') = g_\theta(s) + \gamma h_\omega(s') - h_\omega(s)$ and $D = \frac{e^f}{e^f + \pi_\theta(a | s)}$
 - 4: **discriminator step:** ascend the binary cross-entropy of D on $\mathcal{D}_E$ (label 1) versus $\mathcal{B}$ (label 0), w.r.t. (θ, ω)
 - 5: **policy step:** update π_θ by TRPO/PPO on $\hat{r} = \log D - \log(1 - D) = f_{\theta, \omega} - \log \pi_\theta$
 - 6: **end for**
 - 7: **return** transferable reward g_θ ; policy π_θ ; soft value estimate h_ω
-

Ledger: AIRL

Assumed (Coordinate 1). Everything GAIL assumes, *plus*: the true reward is state-only, and the dynamics are decomposable. In exchange, the shaping ambiguity—the one ambiguity that blocks transfer—is quotiented out by construction.

Approximated (Coordinate 2). As GAIL: samples throughout, no Z , no P . Additionally requires evaluating $\pi(a | s)$ explicitly inside D , so the policy must have a tractable density.

Delivers. $g_\theta \approx r^* + c$, re-optimisable under new dynamics; $h_\omega \approx V_{\text{soft}}^*$, which is a useful by-product in its own right.

Fails when. The reward really depends on actions; the dynamics are not decomposable; or the discriminator is not trained near optimality—the theorem is about the global optimum of a minimax problem, which is not what any optimiser reliably finds.

Part VI

Synthesis

12 Every Method on the Two Coordinates

Table 2: The same four methods, read as answers to “what did you assume?” and “what did you approximate?”

	Max-Margin Abbeel & Ng '04	MaxCausalEnt Ziebart '08/'10	GAIL Ho & Ermon '16	AIRL Fu et al. '18
<i>Coordinate 1: what is assumed</i>				
Behaviour model	expert exactly optimal	Boltzmann-rational, $\pi \propto e^{Q_{\text{soft}}/\beta}$	entropy-regularised optimal	entropy-regularised optimal
Selection principle	largest margin	maximum causal entropy	ψ_{GA} , i.e. min JS on ρ	ψ_{GA} + shaping quotient
Reward class	linear in fixed ϕ	linear in fixed ϕ (or any R_θ)	implicit: discriminator architecture	$g_\theta(s)$ state-only + h_ω shaping
Extra structural assumption	—	—	—	decomposable dynamics
<i>Coordinate 2: what is approximated</i>				
Needs P explicitly?	yes (inner RL + $\mu(\pi)$)	yes: (32) and (35)	no — needs a <i>sampler</i>	no — needs a <i>sampler</i>
Inner expectation via	exact DP / linear solve	exact backward + forward passes	sampled rollouts + classifier	sampled rollouts + classifier
Partition function	absent (no likelihood)	exact, by soft DP	never formed	never formed
Per-iteration cost	QP + full RL solve	$O(T \mathcal{S} ^2 \mathcal{A})$	rollout batch + 2 SGD steps	rollout batch + 2 SGD steps
Scales to continuous control?	no	no (tabular)	yes	yes
<i>Outcome</i>				
Returns	mixed policy + weak w	unique w , soft policy, a likelihood	policy only	policy + transferable g_θ ; $h_\omega \approx V_{\text{soft}}$
Handles noisy experts	poorly	naturally	naturally	naturally
Reward transfers to new dynamics?	only within the class	only within the class	no ($D \rightarrow \frac{1}{2}$)	yes, under Thm. 11.4
Chief failure mode	infeasible constraints; feature design	intractable outside small tabular MDPs	minimax instability; no reward at the end	assumption violation; optimisation never reaches the optimum

The one-paragraph version

Max-margin fixed the ambiguity geometrically and paid with a brittle behaviour model. Maximum entropy replaced the geometry with a likelihood, which handled noise and gave a convex problem, and paid with an intractable partition function. Occupancy measures revealed that both were solving a distribution-matching problem in disguise. GAIL then estimated that distributional discrepancy by classification instead of by integration, which made it scale—at the cost of throwing the reward away entirely. AIRL put a reward back in by constraining the classifier so that a state-only component and a shaping component are forced apart, buying transfer with one more assumption. At no point did anyone

recover a true reward from data alone, and at no point did anyone claim to.

13 The Three Questions, Answered on One Page

Almost every beginner arrives with the same three questions. Having built the machinery, we can now answer them crisply.

Q1. “Everything is finite here—what happens in the real, infinite case?”

The *statistics* are unchanged; only the *computation* changes. Concretely: an infinite horizon needs discounted causal entropy and a soft Bellman fixed point, which still contracts (Lemma 6.3), so nothing is lost. A large or continuous $\mathcal{S}$ destroys the tabular passes and forces function approximation and sampling. A continuous $\mathcal{A}$ makes the soft-max over actions an intractable integral, handled variationally. And on a continuum, the maximum-entropy objective is only well-posed as a *relative* entropy $-\text{KL}(p||q)$ —which is why we insisted on the reference measure q in Definition 5.1 from the very first line, even though in the finite deterministic case it is a harmless constant. The reason to learn IRL discretely first is not that discrete problems matter; it is that in the discrete case every quantity is definable, so you can see precisely which ones later become approximations.

Q2. “Do the features have to be hand-designed?”

No. The maximum-entropy learning rule is $\mathbb{E}_{\text{expert}}[\nabla_{\theta} R_{\theta}] - \mathbb{E}_{\text{model}}[\nabla_{\theta} R_{\theta}]$ (Proposition 7.1) for *any* differentiable reward class; the classical rule $\mu_E - \mathbb{E}[\Phi]$ is the special case $R_{\theta} = w^{\top} \phi$. What a fixed linear ϕ actually bought was a finite sufficient statistic, a convex dual, a class-wide performance guarantee, and interpretability—and what it cost was silent misspecification. Learning ϕ trades convexity for expressiveness and introduces a new $(w, \phi) \sim (A^{-\top} w, A\phi)$ ambiguity. Most importantly, the choice never disappears: in GAIL and AIRL the discriminator architecture *is* the reward class, performed less visibly.

Q3. “What if we don’t know P and have to explore?”

First separate the two jobs. Estimating the *expert’s* statistics needs no model at all—it is an average over the demonstrations. Estimating the *current candidate policy’s* statistics is the only place P was ever used in exact MaxEnt IRL: the successor expectation in the soft backward pass (32), and the visitation propagation in the forward pass (35). Both are expectations, so both can be replaced by sampling whenever you can *draw* transitions, even if you can never write P down. You then pay in variance, sample complexity, coverage, and possibly in the real-world risk of exploring; you do not pay any change in the IRL objective itself. Pushed one step further, the observation that we only ever needed to *compare* two sampled distributions replaces integration by classification—and that step is GAIL.

14 A Checklist for Reading an IRL Paper

1. What is the behaviour model? Exact optimality, Boltzmann rationality with which temperature, or something else?
2. What is the reward class, *including* when it is hidden in an architecture?
3. What is the selection principle, and is it stated or smuggled?

4. Which quantity plays the role of $\mathbb{E}_{\text{model}}[\cdot]$, and how is it obtained—exact DP, rollouts, importance weights, or a discriminator?
5. Does the method need P , a simulator, online interaction, or only offline data?
6. What exactly is claimed to be recovered—a policy, a reward equivalence class, or a specific reward—and under what conditions?
7. If transfer is claimed, which ambiguity was removed to make transfer possible, and by what assumption?

15 Common Errors, Collected

These are mistakes that recur in secondary expositions, including in earlier drafts of these notes. Each is discussed at the reference given.

1. **Deriving $p(\tau) \propto e^{w^\top \Phi(\tau)}$ and later asserting $p(\tau) \propto q(\tau)e^{w^\top \Phi(\tau)}$.** These solve different programs. Put the reference measure into the objective from the start (Definition 5.1).
2. **Writing the maximum-entropy program as $-\int p \log p$ on a continuous trajectory space.** Differential entropy is not reparameterisation-invariant (Section 6).
3. **Using the naive soft backward pass under stochastic dynamics.** It is risk-seeking and implicitly rewrites the transition kernel (Proposition 5.11); use maximum causal entropy (Theorem 5.13).
4. **Attributing potential-based shaping to Ng & Russell (2000).** It is Ng, Harada & Russell (1999), and the converse—that this is the *only* dynamics-independent invariance—is what makes it the right nuisance class.
5. **Returning the last policy from Abbeel & Ng’s algorithm.** The guarantee is for a mixture (Section 4).
6. **Mixing GAIL discriminator conventions, or treating $-\log(1 - D)$, $\log D$ and the log-odds as interchangeable.** They induce different survival biases (Section 9).
7. **Claiming GAIL learns a reward.** At convergence $D \equiv \frac{1}{2}$ (Section 9).
8. **Writing AIRL’s task term as $g_\theta(s, a)$.** The disentanglement theorem requires a state-only g (Section 11).
9. **Confusing uniqueness of V^* with identifiability of R , or uniqueness of the estimator w with identifiability of the estimand.** (Sections 3 and 5.)

16 Conclusion

Inverse reinforcement learning is easy to state and impossible to solve as stated. The forward map from reward to behaviour destroys information, so behaviour can never determine reward; and the natural repairs all require an expectation under the policy that a candidate reward induces, which is a full forward-RL problem nested inside every gradient step.

Reading the field through those two failures makes its history legible. Max-margin and maximum entropy are two answers to the first failure, and maximum entropy wins because it produces a likelihood: noise is modelled rather than excluded, and the estimation problem becomes convex. The exact tabular algorithm we built in Section 5 is the reference implementation of that idea, and its two passes show, with no ambiguity, where a transition model is required. Every

method after it is an answer to the second failure. Occupancy measures reveal what the inner expectation really is; sampling replaces the forward pass; importance weighting reuses samples; and finally the realisation that we only ever needed to *compare* two sampled distributions replaces integration with classification, which is GAIL. AIRL then returns to the first failure with the leverage that adversarial training provides, and buys back exactly one ambiguity—potential shaping—at the price of exactly one assumption—a state-only reward.

The lesson worth carrying away is not any single algorithm. It is that in IRL, every capability is purchased with an assumption, and the assumptions are always visible if you look for them in the right two places.

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